CSC3210 - Quiz - OS Motivation and Computer Architecture

Name: Solution

Answer the questions in the spaces provided. Work is to be done on your own without a computer/phone. Justify your answers and show all work where appropriate.

1. (10 points) What is an abstraction in the context of operating systems? Support your answer with a specific example.

   Solution: An abstraction in operating systems is a way to simplify complex hardware and software interactions, allowing programmers to work without needing to manage low-level details.

   For example, file systems are abstraction provided by the OS that makes sharing and accessing persistent media (hard drives, solid state disks, etc.) easier.

2. (10 points) What is isolation in the context of multiple processes in computing? Why is isolation of processes important? Support your answer with a specific example.

   Solution: Isolation is the separation of one process/application from another to protect memory and other allocated resources. Isolation is important to ensure that different process/applications running on the same system do not interfere with each other.

   For example, a music player should not be able to access the credit card information entered into a web browser.

3. (10 points) What is multiprogramming? Why is it beneficial? Support your answer with a specific example.

   Solution: Multiprogramming is the ability to run multiple programs simultaneously on a computer. It is beneficial because it improves hardware utilization by allowing multiple tasks to share CPU time, making the system more efficient.

   For example, allowing a web browser to run multiple tabs at the same time allows multiple websites to be loaded at approximately the same time.

4. (10 points) Describe the fetch-execute cycle in a CPU? What steps are involved? What does the CPU do at each step?

Solution: The fetch-execute cycle involves fetching an instruction from memory, decoding it, executing it, and then writing back the result. This cycle is repeated continuously by the CPU.

1. Fetch - using the address contained in the program counter, the CPU loads the next instruction from memory
2. Decode - the CPU control unit decodes the instruction to determine what operation needs to be performed
3. Execute - the CPU control unit signals the appropriate hardware in the CPU to perform the necessary operation
4. Write-back - the result is written back either to memory, a register, the program counter, etc. based on the action performed